

Technology Stack

Date	30 September 2026
Team ID	PER206AIFITTRACK
Project Name	AI-FitTrack-API
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	VS Code Thunder Client & REST Clients	Interactive REST API client for request configuration, header management, and real-time response inspection.
2	Backend / Server-Side	Node.js (v20+) & Express.js Framework	Fast asynchronous, event-driven runtime providing modular routing, middleware pipelines, and sub-50ms response latency.
3	Database / Data Storage	MongoDB (v7.x) & Mongoose ODM	Flexible NoSQL document store ideal for fitness workout logs; Mongoose provides schema validation and compound indexing.
4	Cloud / Hosting / Deployment	Local Environment (Port 5000), MongoDB Atlas, Docker	Cross-platform containerization and cloud database availability with automated failover and secure replication.
5	Version Control & Automation	Git & GitHub Repository, test-suite.js	Source code versioning, commit tracking, and automated 59-assertion backend testing suite.
6	Third-Party APIs / Other Tools	Google Gemini 1.5 Flash (@google/genai SDK)	High-speed generative AI model fine-tuned for structured JSON extraction and adaptive personalized fitness coaching.

Application Architectural Characteristics

S.No	Characteristic	Design Justification & Architectural Evidence	Implemented Technology
1	Open-Source Frameworks	Express.js for routing; Mongoose for ODM schema modeling; jsonwebtoken for auth; bcryptjs for hashing; dotenv for environment config.	Express.js, Mongoose, jsonwebtoken, bcryptjs
2	Security Implementations	bcrypt password hashing (10 rounds); HMAC-SHA256 signed JWT tokens; Bearer authMiddleware; Mongoose parameterization preventing NoSQL injection.	bcryptjs, JWT, Bearer Token, .env Isolation
3	Scalable Architecture	Layered MVC architecture decoupling routes, controllers, and models. Stateless REST API design enables horizontal scaling across multiple Node.js workers.	MVC Pattern, Stateless REST API, PM2 cluster readiness
4	High Availability	Resilient MongoDB reconnection handlers, unhandled exception trapping in centralized middleware, and automated error recovery.	MongoDB Auto-Reconnect, Process Error Handlers